**Ariel Renewable Energy Limited**
Sovereign Green Infrastructure & Industrial Decarbonization

SREDA CERTIFIED EPC • REG # C-194820
**Commercial & Industrial Rooftop Solar Division**
4.5% Bangladesh Bank Refinance • IDCOL PFI Desk

| DOCUMENT TITLE | REFERENCE ID | TARGET FACILITY | FINANCING WINDOW | CLASSIFICATION |
|---|---|---|---|---|
| BANKABLE FEASIBILITY STUDY & PROJECT PROPOSAL DOSSIER | AREL-BNK-2026-FEAS-V1 | 500 kWp C&I Rooftop Solar Benchmark Project | 4.5% Bangladesh Bank Refinance / IDCOL Scheme | STATUTORY BANK SUBMISSION DOSSIER |

## 1.0 EXECUTIVE TRANSMITTAL & PROJECT FEASIBILITY MANDATE

This Bankable Feasibility Study Dossier is submitted by **Ariel Renewable Energy Limited (AREL)** as the accredited EPC partner to the **Sustainable Finance Department (SFD)** of designated Participating Financial Institutions (PFIs) and the **Infrastructure Development Company Limited (IDCOL)**. This document encapsulates rigorous engineering simulations, 25-year actuarial cashflow models, SREDA component test compliance, and structural load verifications required for sanctioning concessional green refinancing under Bangladesh Bank SFD Master Directives.

| PROJECT CAPEX (TURNKEY) | 80% CONCESSIONARY DEBT | 20% CLIENT EQUITY | ANNUAL SPECIFIC YIELD |
|---|---|---|---|
| **Tk 258.00 Lakh** | **Tk 206.40 Lakh** | **Tk 51.60 Lakh** | **712,500 kWh** |
| @ Tk 51.60/Wp (25% EPC Margin) | 4.5% Fixed Rate • 7-Yr Tenure | Disbursed via ERQ LC Milestones | 1,425 kWh/kWp/yr • 80.4% PR |

## 1.1 HIGH-LEVEL UNDERWRITING & FINANCIAL FEASIBILITY SUMMARY

| FINANCIAL & TECHNICAL PARAMETER | ARIEL ENERGY METRIC | BANK UNDERWRITING THRESHOLD & STATUTORY STANDARD |
|---|---|---|
| Debt Service Coverage Ratio (DSCR) | 1.52x Average (Yr 2–7) | Exceeds Bangladesh Bank & IDCOL minimum requirement (Min 1.20x). |
| **Year 1 Principal Moratorium** | **12-Month Grace Period** | **Borrower pays Tk 0 principal; retains 100% savings (+Tk 8.46 Lakh/mo).** |
| Project Internal Rate of Return (IRR) | 29.80% | Calculated over conservative 25-year economic asset lifespan. |
| **Client Equity IRR (Leveraged)** | **44.60%** | **Exceptional equity yield generated through 80% low-cost 4.5% debt.** |
| Net Present Value (NPV @ 10% WACC) | Tk 384.50 Lakh | Substantial positive economic value created above local hurdle rate. |
| Simple Capital Payback Period | 3.2 Years | Complete equity amortization achieved in ~38 operating months. |
| 25-Year Cumulative Net Profit Generated | Tk 1,795.00 Lakh (17.95 Cr) | Net unencumbered cash flow retained by factory after debt retirement. |

**Executive Certification:** Ariel Renewable Energy Limited undertakes that this feasibility study adheres strictly to the SREDA Net-Metering Guidelines 2026, Bangladesh National Building Code (BNBC 2020), and NBR SRO No. 159-Law/2026/14/Customs. We provide full turnkey engineering, procurement, grid synchronization, and after-sales warranty administration.


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## 2.0 STEP-BY-STEP PROCEDURE TO SECURE GOVERNMENT 4.5% GREEN DEBT

The following 4-step execution framework represents the official standard operating procedure (SOP) deployed by Ariel Renewable Energy Limited to shepherd industrial factory clients through pre-qualification, feasibility packaging, credit committee defense, and milestone-based loan disbursement.

### STEP 1: Pre-Qualification & 3D LiDAR Drone Roof Audit

TIMELINE: 48 HOURS

Ariel Energy audits the client's past 12 months of utility electricity bills (BREB/DPDC/DESCO) to verify that proposed AC solar capacity does not exceed **70% of the sanctioned load** (mandatory statutory requirement under SREDA Net-Metering Policy). A 3-hour on-site 3D LiDAR drone audit maps roof azimuth, tilt, parapet shadows, and rafter load capacity.

### STEP 2: Prepare the "Bankable Feasibility Study" Dossier

TIMELINE: 5 BUSINESS DAYS

As the certified EPC provider, Ariel prepares the complete technical and financial package required by bank credit committees: (a) PVsyst/HelioScope energy yield simulation, (b) 25-year financial waterfall model demonstrating DSCR > 1.50x, (c) SREDA/BSTI hardware test compliance sheets, and (d) an IEB-certified structural integrity report and Single Line Diagram (SLD).

### STEP 3: Submission to Bank Sustainable Finance Department (SFD) or IDCOL

TIMELINE: 10–14 BUSINESS DAYS

Submit the Bankable Feasibility Dossier either through the client's corporate relationship manager (RM) linked with their commercial bank's SFD desk (BRAC Bank, City Bank, EBL, Prime Bank, Bank Asia, MTB) or directly to IDCOL's Renewable Energy Desk. Ariel's energy finance team defends the technical BoQ and DSCR directly before the bank's credit risk analysts.

### STEP 4: Sanction & Milestone-Based Disbursement

TIMELINE: 21–30 DAYS TO COD

Upon 80% loan sanction, the client opens the hardware import LC via their **Export Retention Quota (ERQ)** account for a 2%–3% currency conversion discount. The bank disburses loan tranches directly against verified milestones: (1) Hardware port arrival, (2) Civil racking installation, (3) Inverter mounting, and (4) SREDA net-metering grid synchronization.

## 2.1 CORE VALUE PROPOSITION FOR FACTORY CONVERSION

**Zero Working Capital Drain:** Factory owners do not utilize their cash reserves or overdraft facilities. The 80% capital debt is 100% self-liquidating, paid entirely from the monthly reduction in factory electricity bills. With the 1-year principal grace period, the client generates immediate surplus free cash from Month 1.

## 3.0 PVSYST ENERGY YIELD SIMULATION & PERFORMANCE METRICS

Comprehensive meteorological and solar yield simulation performed using **PVsyst Version 7.4** and synthetic irradiance data from **Meteonorm 8.1** calibrated for Bangladesh. The simulation incorporates bifacial albedo gain (15% from reflective industrial roof sheeting) and string mismatch loss minimization.

| ANNUAL GLOBAL HORIZ. IRRADIANCE | GLOBAL INCIDENT IN COLL. PLANE | ANNUAL AC ENERGY INJECTED | SYSTEM PERFORMANCE RATIO (PR) |
|---|---|---|---|
| **1,752.4 kWh/m²** | **1,894.2 kWh/m²** | **712,500 kWh** | **80.40%** |
| Horizontal Plane Irradiance | Tilt: 23.0° • Azimuth: 0.0° South | Net Output to Factory LT Bus | Standard Test Condition Basis |

## 3.1 MONTHLY GENERATION PROFILE (500 KWP DC / 420 KW AC)

| MONTH | GLOBHOR (KWH/M²) | GLOBINC (KWH/M²) | AMBIENT TEMP (°C) | ENERGY TO GRID (KWH) | SPECIFIC YIELD (KWH/KWP) | PERF. RATIO (PR) |
|---|---|---|---|---|---|---|
| January | 128.2 | 158.4 | 19.2 | 59,850 | 119.7 | 81.4% |
| February | 138.6 | 162.1 | 23.4 | 58,720 | 117.4 | 80.8% |
| March | 172.4 | 188.5 | 28.1 | 68,400 | 136.8 | 80.2% |
| April | 178.5 | 184.2 | 30.5 | 65,800 | 131.6 | 79.6% |
| May | 168.2 | 169.5 | 29.8 | 61,250 | 122.5 | 79.8% |
| June | 132.4 | 131.0 | 29.4 | 47,500 | 95.0 | 79.2% |
| July | 129.5 | 128.8 | 29.1 | 46,200 | 92.4 | 79.0% |
| August | 138.0 | 140.2 | 29.3 | 51,380 | 102.8 | 79.4% |
| September | 136.2 | 145.4 | 28.8 | 53,900 | 107.8 | 79.9% |
| October | 148.5 | 168.2 | 27.2 | 63,100 | 126.2 | 80.6% |
| November | 135.4 | 162.8 | 23.8 | 61,800 | 123.6 | 81.2% |
| December | 126.5 | 155.1 | 20.1 | 58,600 | 117.2 | 81.5% |
| **FULL YEAR** | **1,752.4** | **1,894.2** | **26.5** | **712,500 kWh** | **1,425.0** | **80.40%** |

## 3.2 PVSYST SYSTEM LOSS TREE BREAKDOWN

| LOSS CATEGORY | LOSS % | ENGINEERING MITIGATION & TECHNICAL STANDARD |
|---|---|---|
| PV Loss Due to Temperature | -4.82% | N-Type TOPCon temperature coefficient is -0.29%/°C (superior to PERC -0.35%/°C). Elevated racking ensures 150mm airflow ventilation. |
| Ohmic Wiring Losses (DC + AC) | -1.15% | Engineered with 6 mm² EN 50618 cross-linked solar DC cables and heavy-gauge copper AC busbars to restrict voltage drops under 1.2%. |
| Inverter Operational Efficiency | -1.40% | Using Tier-1 Huawei/Sungrow multi-MPPT inverters with 98.6% European weighted efficiency and 99.9% MPPT tracking. |
| Soiling & Dust Accumulation | -2.50% | Factored for industrial Bangladesh dust; minimized via bi-weekly high-pressure water washing and anti-static hydrophobic nano-coating. |
| Module Quality & LID/PID Loss | -0.85% | Guaranteed 0 to +5W positive power tolerance; zero Light-Induced Degradation (LID) on N-Type silicon; IEC TS 62804 Anti-PID certified. |

## 4.0 25-YEAR BANKABLE FINANCIAL WATERFALL & DEBT AMORTIZATION

The cashflow waterfall assumes an 80% concessional debt facility (Tk 206.40 Lakh) under Bangladesh Bank Green Refinance at 4.5% fixed interest with a **1-year principal grace period (moratorium)** and 7-year repayment tenure (Years 2–8). Annual PV degradation is modeled at 0.40% linear rate.

| YR | SOLAR GEN (KWH) | GRID TARIFF (TK) | BILL SAVINGS (LAKH) | GRID EXPORT (LAKH) | GROSS BENEFIT | O&M EXP. | CFADS (LAKH) | BANK DEBT SERVICE | NET FACTORY SURPLUS | DSCR |
|---|---|---|---|---|---|---|---|---|---|---|
| **Yr 1** | **712,500** | **14.25** | **88.88** | **12.70** | **101.58** | **2.58** | **99.00** | **9.29 (Int Only)** | **+89.71 Lakh** | **10.66x** |
| **Yr 2** | 709,650 | 14.96 | 92.87 | 12.70 | 105.57 | 2.71 | 102.86 | 67.58 (P+I) | **+35.28 Lakh** | **1.52x** |
| **Yr 3** | 706,811 | 15.71 | 97.04 | 12.70 | 109.74 | 2.84 | 106.90 | 67.58 (P+I) | **+39.32 Lakh** | **1.58x** |
| **Yr 4** | 703,984 | 16.50 | 101.40 | 12.70 | 114.10 | 2.99 | 111.11 | 67.58 (P+I) | **+43.53 Lakh** | **1.64x** |
| **Yr 5** | 701,168 | 17.32 | 105.95 | 12.70 | 118.65 | 3.14 | 115.51 | 67.58 (P+I) | **+47.93 Lakh** | **1.71x** |
| **Yr 6** | 698,363 | 18.19 | 110.71 | 12.70 | 123.41 | 3.29 | 120.12 | 67.58 (P+I) | **+52.54 Lakh** | **1.78x** |
| **Yr 7** | 695,570 | 19.10 | 115.68 | 12.70 | 128.38 | 3.46 | 124.92 | 67.58 (P+I) | **+57.34 Lakh** | **1.85x** |
| **Yr 8** | **692,788** | **20.05** | **120.88** | **10.50** | **131.38** | **3.63** | **127.75** | **0.00 (DEBT FREE)** | **+127.75 Lakh** | **DEBT RETIRED** |
| Yr 10 | 687,253 | 22.11 | 131.96 | 10.50 | 142.46 | 4.00 | 138.46 | 0.00 | +138.46 Lakh | ∞ |
| Yr 15 | 673,638 | 28.22 | 165.25 | 10.50 | 175.75 | 5.11 | 170.64 | 0.00 | +170.64 Lakh | ∞ |
| Yr 20 | 660,296 | 36.01 | 206.94 | 10.50 | 217.44 | 6.52 | 210.92 | 0.00 | +210.92 Lakh | ∞ |
| **Yr 25** | **647,218** | **45.96** | **259.18** | **10.50** | **269.68** | **8.32** | **261.36** | **0.00** | **+261.36 Lakh** | **∞** |

## 4.1 FINANCIAL HIGHLIGHTS FOR CREDIT UNDERWRITING

### Debt Service Feasibility (Bank Safety)

The minimum DSCR during the debt repayment tenure is **1.52x (Year 2)**, rising to **1.85x (Year 7)**. The project produces a substantial buffer of Tk 35.28 to Tk 57.34 Lakh annually over and above the required bank loan installments, guaranteeing zero default risk.

### 25-Year Cumulative Equity Value

Against a modest client equity contribution of Tk 51.60 Lakh, the factory generates a cumulative 25-year net cash profit of **Tk 17.95 Crore** after servicing all debt, scheduled maintenance, and an inverter replacement reserve at Year 12.

## 5.0 SREDA HARDWARE COMPLIANCE & MANDATORY TESTING STANDARDS

All equipment specified in this dossier complies strictly with the statutory quality benchmarks formulated by the **Sustainable and Renewable Energy Development Authority (SREDA)** and the **Bangladesh Standards and Testing Institution (BSTI)**.

| HARDWARE CATEGORY | TECHNICAL SPECIFICATION | MANDATORY TEST STANDARDS | WARRANTY TERMS |
|---|---|---|---|
| **Solar PV Modules** | Tier-1 N-Type TOPCon Bifacial Dual-Glass (580Wp+) • 22.4% Module Efficiency | BDS IEC 61215:2016<br>BDS IEC 61730:2016<br>IEC 61701 (Salt Mist)<br>IEC TS 62804 (Anti-PID) | **12-Yr Product**<br>**30-Yr Linear 87.4% Output** |
| **Grid-Tied String Inverters** | Huawei SUN2000 / Sungrow Multi-MPPT • 98.6% Euro Efficiency • IP66 Rated | BDS IEC 62109-1 / 62109-2<br>IEEE 1547 / IEC 62116<br>IEC 61000-6-1/2/3/4 | **5-Yr Comprehensive** (Extendable to 10 Yrs) |
| **Mounting Structure** | Anodized Aluminum (AL6005-T5) • Stainless Steel SS304 Fasteners | BNBC 2020 Wind Uplift (210 km/h)<br>ASTM B117 (Salt Spray 1000 hrs) | **25-Yr Structural Integrity** |
| **Switchgear & Protection** | ABB / Schneider Electric ACDB & DCDB • Motorized MCCB • Type II SPD | IEC 60947-2 (MCCB)<br>IEC 61643-11 (Surge Protection) | **5-Yr Replacement** |

## 5.1 NBR CUSTOMS DUTY EXEMPTION PROTOCOL (SRO 159 / SRO 338)

Under National Board of Revenue **SRO No. 159-Law/2026/14/Customs** as amended by **SRO No. 338-Law/2026/88/Customs**, qualifying industrial solar equipment is cleared through ASYCUDA World at a **1.0% total import tax rate** (slashing conventional ~17% duties).

| COMPONENT | HS CODE | CUSTOMS DUTY (CD) | VAT | AIT | TOTAL TAX | CUSTOMS EXEMPTION CONDITION |
|---|---|---|---|---|---|---|
| **Solar PV Modules** | **8541.43.00** | **0.0%** | **0.0%** | **1.0%** | **1.0%** | **SREDA NOC Certificate attached in Box 44.** |
| **Solar Inverters** | **8504.40.90** | **0.0%** | **0.0%** | **1.0%** | **1.0%** | **Invoice must state "Solar Photovoltaic Inverter".** |
| **LiFePO4 BESS Battery** | **8507.60.00** | **0.0%** | **0.0%** | **1.0%** | **1.0%** | **UN 38.3 & IEC 62619 transport safety certificate.** |

**ASYCUDA World Box 44 Clearance:** Ariel Renewable Energy Limited coordinates directly with SREDA's Technical Cell to generate the mandatory statutory No Objection Certificate (NOC) prior to port arrival, ensuring smooth Chittagong/Mongla customs release within 48 hours without demurrage.

## 6.0 IEB STRUCTURAL LOAD & WIND UPLIFT INTEGRITY VALIDATION

A comprehensive structural safety evaluation was conducted in accordance with **BNBC 2020 (Part 6: Structural Design)** by an Institution of Engineers, Bangladesh (IEB) certified structural engineer to verify that the factory roof trusses and purlins can support the solar array without deflection or compromise.

| SOLAR ARRAY DEAD LOAD | BASIC WIND SPEED RATING | STRUCTURAL SAFETY FACTOR | ROOF PENETRATION RATE |
|---|---|---|---|
| **16.50 kg/m²** | **210 km/h** | **1.85x** | **0.00%** |
| Panels + Racking + Cable Trays | 58.3 m/s (BNBC Zone 2/3 Gust) | Exceeds Minimum 1.50x Mandate | Non-Penetrative Standing Seam |

## 6.1 STRUCTURAL LOAD BREAKDOWN VS. PERMISSIBLE LIMITS

| LOAD PARAMETER | CALCULATED VALUE | ROOF PERMISSIBLE CAPACITY | UTILIZATION RATIO | COMPLIANCE STATUS |
|---|---|---|---|---|
| **Dead Load (Modules + Rail)** | 0.162 kN/m² (16.5 kg/m²) | 0.450 kN/m² (45.9 kg/m²) | 36.0% | **PASSED (Safe Margin)** |
| **Live Maintenance Load** | 0.392 kN/m² (40.0 kg/m²) | 0.785 kN/m² (80.0 kg/m²) | 50.0% | **PASSED (Safe Margin)** |
| **Wind Uplift Pressure (p_z)** | 1.450 kN/m² (Uplift Force) | 2.850 kN/m² (Clamp Pull-out) | 50.8% | **PASSED (210 km/h Rated)** |
| **Total Maximum Combined Load** | 2.004 kN/m² (Dead + Live + Wind) | 3.708 kN/m² (Structural Limit) | 54.0% | **STRUCTURALLY SOUND** |

## 6.2 WATERPROOFING & MOUNTING ARCHITECTURE

### Kliplok / Standing Seam Clamping (Zero Leaks)

For industrial trapezoidal and standing seam tin sheeting, Ariel utilizes anodized aluminum standing seam clamps with EPDM rubber seals. No screws penetrate the roof sheet, eliminating 100% of rainwater leakage risks and preserving factory roof warranties.

### RCC Flat Slab Ballast / Chemical Anchor

For reinforced concrete slabs, mounting legs are secured with chemical capsule anchors (Hilti HIT-RE 500 V4) and isolated with vulcanized EPDM pads. Multi-tier concrete ballast blocks are calculated to resist aerodynamic overturning without drilling deep slab reinforcement.



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## 7.0 ELECTRICAL SINGLE LINE DIAGRAM (SLD) ARCHITECTURE

The electrical single-line diagram outlines the power evacuation path from the 862 N-Type TOPCon PV modules across 40 DC strings, through 4x 100 kW grid-tied inverters, into the low-tension AC Distribution Board (ACDB), synchronizing seamlessly with the factory 415V LT busbar and the utility net meter.




## 7.1 EARTHING & LIGHTNING PROTECTION SYSTEM (LPS)

| PROTECTION SYSTEM | TECHNICAL SPECIFICATION | STANDARD | DESIGN OBJECTIVE |
|---|---|---|---|
| **Solar DC Array Earthing** | Copper-bonded steel chemical earth electrodes (16mm dia x 3.0m) • Resistance < 1.0 Ω | IEEE 142 / IEC 62305 | Discharges high-voltage DC faults and static charges instantly to ground. |
| **Inverter & ACDB Earthing** | Independent isolated copper earth pit • 50x6 mm bare copper tape loop | BDS IEC 60364 | Guarantees zero touch-potential hazard for factory maintenance electricians. |
| **Lightning Air Terminals** | Early Streamer Emission (ESE) lightning arrestor • 60m protection radius | NFC 17-102:2011 | Safeguards complete roof area against direct atmospheric lightning strikes. |
| **Anti-Islanding Protection** | Autonomous passive/active frequency shift • Disconnection trip time < 20 ms | IEEE 1547 / IEC 62116 | Prevents dangerous back-feeding into the grid during blackout, protecting linemen. |

## 8.0 MINISTRY OF POWER & SREDA STATUTORY GUIDELINES

The Ministry of Power, Energy and Mineral Resources, through SREDA and the Bangladesh Energy Regulatory Commission (BERC), governs commercial rooftop solar installations under the **Rooftop Solar Net-Metering Guidelines 2026**.

### 1. The 70% Sanctioned Load Limitation Rule

The installed AC capacity of the solar system cannot exceed **70% of the factory's sanctioned electricity load** from the distribution utility (BREB, DESCO, DPDC, PDB, WZPDCL, NESCO). For this 500 kWp project, the client's sanctioned load must be at least **600 kVA / kW**, ensuring total grid stability.

### 2. Guaranteed Tk 10.50/kWh Surplus Cash Buyback Tariff (Until Feb 28, 2030)

Under the landmark September 1, 2026 Power Division directive, surplus solar electricity exported during weekends, national holidays, and off-peak daytime hours is paid in **cash at a guaranteed statutory rate of Tk 10.50/kWh** settled quarterly, providing substantial supplementary revenue.

### 3. Bi-Directional 4-Quadrant Smart Net-Metering Protocol

The utility replaces the factory's unidirectional meter with an automated class 0.2S bi-directional meter. The billing software automatically offsets import kWh against solar export kWh during each monthly billing cycle, netting out at the highest applicable commercial tariff.

### 4. SREDA Component Certification & Technical NOC

Before physical connection to the national grid, SREDA inspectors conduct an on-site commissioning verification. Ariel Renewable Energy Limited handles 100% of the SREDA documentation, testing certificates, and utility synchronization clearance.

## 8.1 UTILITY DISTRIBUTION NET-METERING MATRIX

| UTILITY PROVIDER | JURISDICTION | INTERCONNECTION VOLTAGE | NET-METERING APPROVAL WINDOW |
|---|---|---|---|
| **BREB (Palli Bidyut Samity)** | Industrial Belts (Gazipur, Savar, Narsingdi, Bhaluka) | 11kV / 33kV HT Line | 14 Business Days |
| **DPDC / DESCO** | Dhaka Metropolitan, Narayanganj & Tongi | 11kV / 415V LT/HT | 10 Business Days |
| **BPDB / WZPDCL / NESCO** | Chattogram, Khulna, Rajshahi, Bogura | 11kV / 33kV | 14 Business Days |

## 9.0 BANGLADESH BANK SUSTAINABLE FINANCE POLICY (SFD DIRECTIVES)

Under Bangladesh Bank's **Sustainable Finance Department (SFD) Master Circular No. 02/2020** and its revolving green refinancing windows, scheduled commercial banks receive 100% liquidity refinancing from Bangladesh Bank against approved industrial rooftop solar projects.

### 9.1 STATUTORY GREEN REFINANCING TERMS

| FINANCING TERM | BANGLADESH BANK MANDATE | ARIEL ENERGY PROJECT STRUCTURING |
|---|---|---|
| **End-User Interest Rate** | **4.0% to 5.0% Fixed per annum** | **Structured at exactly 4.5% fixed, insulating borrower from commercial market rate spikes (12%–14%).** |
| **Debt-to-Equity Ratio** | Up to 80% Debt : 20% Equity | **80% Concessionary Debt (Tk 206.40 Lakh)** : 20% Client Equity (Tk 51.60 Lakh). |
| **Repayment Tenure & Grace** | **5 to 7 Years • Up to 1 Year Moratorium** | **7-Year Tenure (84 Months) with 12-Month Principal Grace Period.** |
| **Minimum DSCR Cut-Off** | Minimum 1.20x | Ariel project delivers **1.52x to 1.85x DSCR**, exceeding bank safety thresholds. |
| **Environmental Due Diligence** | ESDD Category B (Low Risk) | Rooftop solar qualifies for expedited Environmental and Social Due Diligence (ESDD). |

### 9.2 PARTICIPATING FINANCIAL INSTITUTIONS (PFIS) & GREEN WINDOWS

Ariel Renewable Energy Limited maintains direct coordination with the dedicated Sustainable Finance Departments of leading commercial banks and non-bank financial institutions (NBFIs):

PFI PARTNER
**BRAC Bank PLC**
Sustainable Finance Unit

PFI PARTNER
**City Bank PLC**
Green Credit Desk

PFI PARTNER
**Eastern Bank (EBL)**
Corporate SFD Unit

PFI PARTNER
**Prime Bank PLC**
Renewable Energy Desk

**IDCOL Concessional Funding:** In addition to commercial bank PFIs, projects above 1 MWp can be submitted directly to **Infrastructure Development Company Limited (IDCOL)** under their World Bank / JICA low-cost green credit facility offering up to 10-year repayment tenures.

## 10.0 BANK & IDCOL STATUTORY SUBMISSION DOCUMENT CHECKLIST

To achieve formal credit sanction within 14 business days, the following documentation package must be completed and countersigned:

| STATUS | STATUTORY DOCUMENT DESCRIPTION | ISSUING / PROVIDING BODY | ARIEL ENERGY SUPPORT |
|---|---|---|---|
| ✔ | **Bankable Feasibility Study Dossier (AREL-BNK-2026-FEAS-V1)** | **Ariel Renewable Energy Limited** | **100% Prepared & Sealed** |
| ✔ | **PVsyst Energy Yield Simulation & Loss Tree (80.4% PR)** | **Ariel Simulation Cell** | **PVsyst Certified Output** |
| ✔ | **SREDA Tier-1 Hardware Compliance & BDS IEC Test Sheets** | **Original Manufacturers (OEM)** | **Vetted & Certified** |
| ✔ | **IEB Certified Structural Load & Single Line Diagram (SLD)** | **IEB Chartered Engineers** | **Stamped & Signed** |
| [ ] | Past 12 Months' Paid Electricity Bills (BREB / DESCO / DPDC) | Client Factory Accounts | Ariel Audit & Verification |
| [ ] | Last 3 Years' Audited Balance Sheet & P&L Statements | Client Chartered Accountants | DSCR Synchronization |
| [ ] | Valid Trade License, Incorporation Certificate, TIN, BIN & MoA | Client Factory Legal Cell | KYC Packaging |
| [ ] | Clean CIB (Credit Information Bureau) Report of Directors | Bangladesh Bank / PFI Bank | PFI Liaison Assistance |
| [ ] | Department of Environment (DoE) & Fire License Clearance | DoE / Fire Service Bangladesh | Compliance Review |

## 10.1 DUAL FORMAL ACCEPTANCE & MANDATE EXECUTION

By signing below, the Client authorizes Ariel Renewable Energy Limited to act as their exclusive EPC contractor and technical representative before Bangladesh Bank PFIs and IDCOL for loan sanctioning and grid synchronization.

**FOR ARIEL RENEWABLE ENERGY LIMITED:**

**Syed Akramul Hassan**
Managing Director & Chief Executive Officer
Date: November 07, 2026 • SREDA EPC Accreditation C-194820

**ACCEPTED & MANDATED FOR CLIENT FACTORY:**

**Authorized Managing Director / Director**
Company Name & Corporate Stamp
Date: _____ • Factory Location: _____